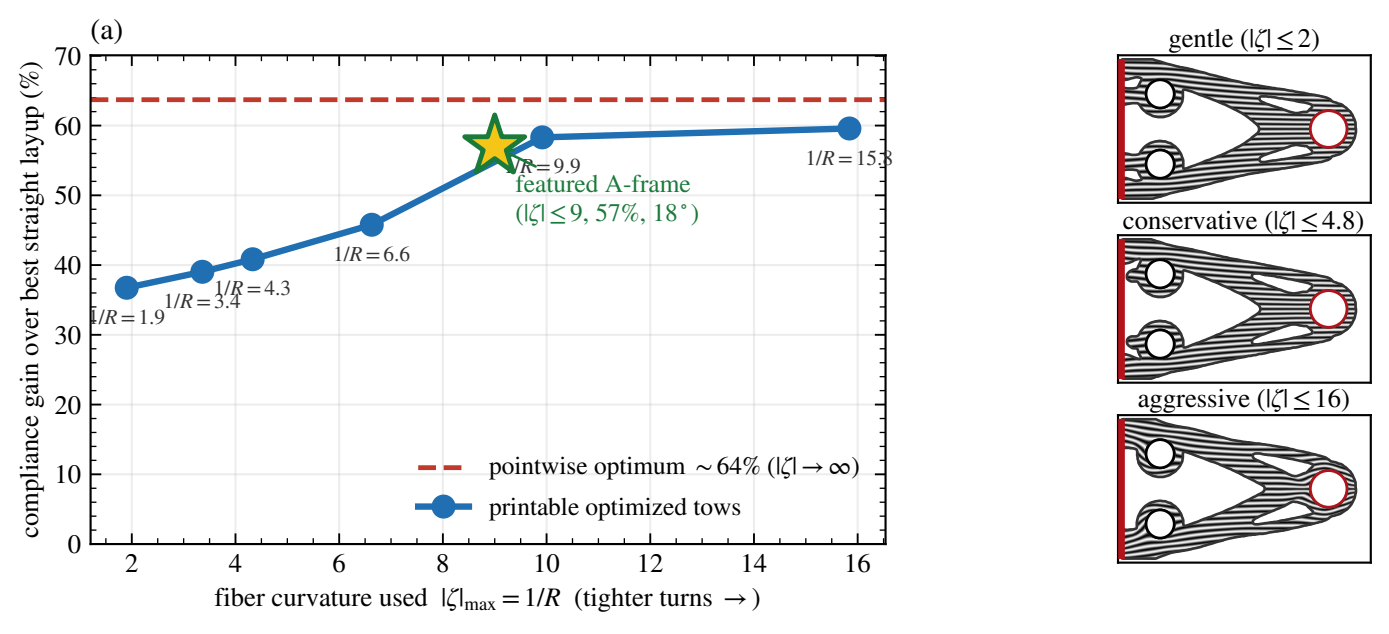

(a)
compliance gain over best straight layup (%)
fiber curvature used $|\zeta|_{\max} = 1/R$ (tighter turns →)
pointwise optimum ~64% ($|\zeta| \to \infty$)
printable optimized tows
1/R = 1.9
1/R = 3.4
1/R = 4.3
1/R = 6.6
1/R = 9.9
featured A-frame
($|\zeta| \le 9$, 57%, 18°)
1/R = 15.3
gentle ($|\zeta| \le 2$)
conservative ($|\zeta| \le 4.8$)
aggressive ($|\zeta| \le 16$)